Simulation project 2024

Group 10

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Team project · collaborators withheld

1. Abstract

1.1 Objective:

To solve two optimisation problems using stochastic approximation methods to find the optimal strategy or parameters.

1.2 Methods:

In project 1, it is to determine the optimal investment allocation vector (p1, p2, p3) to maximise the Sharpe ratio using SPSA. This involves the evaluation of investment return and risk.

In project 2, it is to determine the optimal average service time θ using SPSA to satisfy service quality requirements and minimise costs. This involves quantifying and controlling customer waiting times.

1.3 Key findings:

Stochastic approximation methods are effective in approximating the optimal solution, especially when faced with complex optimisation problems.

In Project 1, stochastic approximation helps to find a balance between multiple investment choices to achieve the best combination of risk and return.

In Project 2, stochastic approximation helps in adjusting the service time to ensure the quality of service and control the cost.

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2. Introduction

2.1 Research Problems

In this study, we are confronted with two projects that both take place in the context of a rapidly changing modern economy and technology. Project 1 addresses the complexity and interdependence of capital markets, particularly in the context of globalisation and rapid advances in information technology. For example, the 2020 New Crown epidemic has had a huge impact on the global economy and increased stock market volatility, requiring investors to adopt more refined strategies to deal with market uncertainty. Our main objective is to help investors in this complex and volatile market environment to develop investment strategies that can deliver solid returns amidst uncertainty by comprehensively analysing market dynamics, company fundamentals and macroeconomic factors.

Project 2 focuses on optimising service station operations in the context of digital service penetration. As customer demand for immediate service and personalised experiences grows, service stations need to improve efficiency and responsiveness while maintaining a human touch. Our goal is to help service stations accurately determine the optimal service time allocation to meet customer expectations and minimise operational costs, while finding a balance between cost control and efficiency improvement.

Both projects are based on a number of necessary assumptions: for Project 1, we assume that investors have access to accurate information about market volatility, company-specific risks, and macroeconomic indicators; and for Project 2, we assume that service stations are able to accurately measure and control service time, and to adjust service processes and resource allocation to affect customer waiting time and overall service quality.

Through these projects, we aim to provide investors and service station managers with scientific and rational decision support, enabling them to make more precise strategic choices in a rapidly changing global economic environment.

2.2 Purpose of the Study

The aim of this research is to apply stochastic approximation (SA) algorithms to provide precise solutions to two real-world problems in the capital market and the service industry. In the capital market, we aim to develop an optimal investment strategy to cope with the current global economic uncertainty and market volatility, with the goal of maximising the Sharpe ratio; while in the service industry, we focus on optimising the customer service process to meet the digital transformation and rising customer expectations, seeking to achieve the best balance between cost control and service quality. Through these studies, we expect to provide investors and service station managers with scientific and rational decision support to help them achieve their goals in a complex and volatile environment.

3. Description of optimization approach

3.1 Algorithm selection

In this study, we chose the Simultaneous Perturbed Stochastic Approximation (SPSA) algorithm to solve the optimisation problems in Project 1 and Project 2, mainly because the SPSA algorithm has a significant advantage when facing complex systems. SPSA does not require explicit derivatives or gradient information of the objective function, which is particularly important in cases such as the GI/GI/1 cohort model in Project 2, where it is difficult to obtain the gradient expression directly. is particularly important. In addition, SPSA estimates the gradient by perturbing all parameters simultaneously, a method that is not only suitable for cases where the gradient is difficult to compute, but also typically provides a lower noise level compared to stochastic gradient descent (SGD) algorithms that perturb each parameter individually, resulting in more stability and accuracy in gradient estimation.

Another advantage of the SPSA algorithm is that it only needs to evaluate two values of the objective function in each iteration, i.e., the results of the positive and negative perturbations, in contrast to traditional SGD algorithms, which may need to be evaluated several times in order to obtain an accurate gradient estimate. This makes SPSA more computationally frugal, especially if the objective function evaluation is costly. In addition, the SPSA algorithm also automatically adjusts the step size during the iteration process, which helps the algorithm to perform a more detailed search when it is close to the optimal solution and converge quickly

when it is far from the optimal solution.

In summary, the SPSA algorithm is chosen for its efficiency and stability in dealing with complex system optimisation problems, and it provides an effective solution to our research problem by reducing the computational cost and improving the accuracy of gradient estimation.

3.2 Well-posedness

In mathematics and engineering, a well-posed problem is a problem that satisfies three conditions: existence (there exists at least one solution), uniqueness (the solution is unique), and stability (the solution varies continuously with the initial conditions of the problem). The following fitness analyses are presented for Project 1 and Project 2:

Project 1: Investment Strategy Optimisation

i. **Existence:** the investment strategy optimisation problem has at least one feasible solution because there are multiple possible ways of allocating capital under the constraint that the total capital allocation is 1.

ii. **Uniqueness:** although there may be multiple solutions, the objective function of maximising the Sharpe ratio helps us to find the optimal solution under the given constraints, which is usually unique or at least locally unique.

iii. **Stability:** the stability of the solution can be ensured by continuously varying constraints and objective functions. The investment allocation strategy reacts sensitively to small changes in market conditions, but the continuity of the objective function and the clarity of the constraints ensure the problem's fitness.

Project 2: Service station queuing optimisation

iv. **Existence:** there exists a solution to the service station queuing optimisation problem because it is always possible to find at least one service time configuration that satisfies the customer's waiting time requirements.

v. **Uniqueness:** the optimal service time configuration may be unique, especially when the objective function (minimising the squared difference of the mean service time) reaches a global minimum under specific constraints.

vi. **Stability:** the stability of the problem is ensured by the continuous response of the service time configuration to customer arrival patterns and service quality requirements. The optimisation model provides stable solutions even under small changes in customer traffic or service demand.

3.3 Coercivity

In mathematical optimisation, coercivity is a key concept that refers to the property of a function that tends to infinity or infinity as the independent variables change. This property is crucial to ensure the existence of a solution to the optimisation problem and the convergence of the algorithm. The following are specific analyses for the mandatory in Project 1 and Project 2:

3.3.1 Mandatory Analysis:

Project 1: Investment Strategy Optimisation

In an investment problem, the coercivity of the vector field may be related to the distributional properties of the investment returns. If the investment ratio pi tends to 0 or 1, the variance of the expected return may tend to 0. This implies that extreme investment strategies (i.e., extreme allocations of capital) will lead to lower risk, although they may also imply lower expected returns.

Coercion here helps the algorithm to identify investment strategies that do not significantly increase the overall risk, thus facilitating the convergence of the algorithm to a solution with balanced risk and return.

Project 2: Service Station Queuing Optimisation

For the distribution of service times, coercion is reflected in the fact that as the service time increases, the average waiting time of customers also increases. This is a natural phenomenon because longer service times cause customers to wait longer in the system. This mandatory helps the optimisation algorithm to find the equilibrium of service time, i.e., to minimise the service time in order to reduce the average waiting time of customers without sacrificing customer satisfaction.

3.4 Gain Size

In stochastic approximation algorithms (SPSA), the choice of gain size or step size is critical to the performance of the algorithm, and it directly affects the search efficiency and convergence of the algorithm. The following is an analysis of the decreasing step size and fixed step size strategies, and the reasons why decreasing step size is usually the better choice:

3.4.1 Decreasing Step Size

Decreasing step size allows the algorithm to take a larger step size initially, which helps to explore the solution space quickly, moving away from local minima and increasing the chances of finding a globally optimal solution. As the algorithm iterates, decreasing the step size can help the algorithm make finer adjustments as it approaches the optimal solution, improving the accuracy of the solution. Moreover, the decreasing step size is adaptive and can dynamically adjust the search strategy according to the current position of the algorithm, balancing the needs of search and refinement.

3.4.2 Fixed step size

Fixed step size can speed up the convergence of the algorithm because the step size remains constant throughout the iteration process, which can reduce the number of iterations in some cases. However, the fixed step size needs to be chosen carefully because if the step size is too large, the algorithm may skip the optimal solution or oscillate in the solution space; if the step size is too small, the algorithm's search efficiency may be reduced. Therefore, a fixed step size may be suitable for cases where the solution space is relatively flat or the algorithm is already close to the optimal solution, but it may not be the best choice in the initial search phase.

3.4.3 Selection

Taking all considerations into account, decreasing step size is usually more superior in SPSA algorithms because it improves the probability of finding a globally optimal solution. It not only allows the algorithm to perform extensive search at an early stage, which helps the algorithm to efficiently approximate the globally optimal solution throughout the iteration, but also avoids falling into the locally optimal solution too early by gradually decreasing the step size.

4. Description of problem

4.1 Investment Strategy Optimisation

The investor is faced with a multi-asset investment problem that requires the allocation of capital among three companies with a view to achieving optimal risk-adjusted returns over a given time horizon. The problem involves random variables and probability distributions, so traditional optimisation methods may not be applicable. The objective function of Project 1 involves random variables, i.e. firms' market capitalisation Xi and returns Yi , which are stochastic.

In our algorithm, use ak = a / (k + 1 + A)**alpha , where k represents the number of iterations.

Project Analysis:

1. **Market value model:** the market value Xi of each firm is given by $Xi = \frac{\rho V + \sqrt{1-\rho^2}\eta i}{max(W,1)}$ where V is a standard normal distribution, ηi is a normal distribution with mean 0 and variance i, and W is an exponential distribution with parameter $\frac{1}{0.3}$. ρ is a weighting factor, which is set to 0.6 here.
2. **return on investment:** if $Xi \geq xi$, the return to firm i is Yi . Returns Yi are independently and uniformly distributed on $[0, Xi]$.
3. **Investment Strategy:** The investor allocates pi proportion of capital to firm i. The expected return is: $piE[Yi \mid Xi \geq xi]$
4. **Sharpe Ratio:** The optimisation objective is to maximise the Sharpe ratio, i.e.: $\max_{p1,p2,p3\geq 0, \sum pi=1} E[\frac{\sum_{i=1}^{3} piYi \mid Xi \geq xi}{std(\sum_{i=1}^{3} piYi \mid Xi \geq xi)}]$

4.2 Service station queuing model optimisation

Customer arrivals and departures at a service station follow a specific stochastic process with the objective of minimising the average customer waiting time while controlling costs, which is a typical queue optimisation problem involving service time and customer satisfaction.

Project Analysis

1. **Waiting Time Recursion:** The waiting time of a customer is given by the following recursive formula: $W_{n+1}(\theta) = max(0, W_n(\theta) + S_n(\theta) - A_{n+1})$ where An is the customer's arrival time, and $S_n(\theta)$ is the service time, dependent on the parameter θ.
2. **Average waiting time:** is the sum of the waiting times of the first N customers divided by N: $W^N(\theta) = \frac{1}{N}\sum_{n=1}^{N} W_N(\theta)$
3. **service quality control:** the objective is to find the minimum θ such that 90% of the customers' average waiting time does not exceed z: $\min_{0<\theta} E[(q_\theta - z)^2]$ where q_θ is the 90% quantile of $W_N(\theta)$.

5. Description of program

5.1 Software and tools used:

- Programming language: ***Python***
- Libraries and frameworks:
    1. numpy: for efficient multi-dimensional array manipulation and numeric counting, generating random numbers.
    2. matplotlib.pyplot: for data visualisation, plotting graphs and charts.

5.2 Programme structure and logic:

5.2.1 Project 1

1. Overview of the programme

1. **import module:**

a. numpy For numerical computation.
b. matplotlib.pyplot is used for data visualisation.
c. random is used to generate random numbers.

2. **Parameter initialisation:**

   Set the number of companies, sample size, weight factor, threshold array, perturbation coefficient and initial investment ratio.

3. **random variable generation:**

   Use NumPy to generate three key random variables: common factors V (standard normal distribution), market shocks W (exponential distribution) and firm-specific risk eta (normal distribution).

4. **Calculate the company market value:**

   Calculate the market value X of each firm at each point in time according to the given formula.

5. **Calculate the return on investment:**

   Calculate the possible return on investment Y for each company based on the market value and the threshold.

6. **Define the Sharpe ratio function:**

Implement a function sharpe_ratio to calculate the Sharpe ratio of a portfolio for a given investment ratio.

7. **Define the SPSA optimisation function:**
   a. Implement the spsa_optimize function for the SPSA algorithm, including: initialisation parameters and history.
   b. iterative process, including the generation of perturbation vectors, the calculation of Sharpe ratios before and after the perturbation, gradient estimation, and the update of the investment ratio.
   c. ensure the constraints of investment ratio (non-negative and sum to 1).
      i. In detail, when the sum of three fractions is more than 1, let them divided by thier norm respectively to reach the constrain.
      ii. Also, we pick the maximum between each value and zero to ensure all non-negative
   d. record the Sharpe ratio and investment ratio for each iteration.

8. **Set the SPSA algorithm parameters:**

   Define the step size parameter, decay parameter and number of iterations of the SPSA algorithm to ensure that the investment ratio can be converged.

9. **Execute SPSA optimisation:**

   Call the spsa_optimize function and pass the relevant parameters to perform the optimisation process.

10. **Visualisation of results:**

    Use Matplotlib to plot the trend of Sharpe ratio and investment ratio with the number of iterations.

11. **Output the optimal solution:**

    Print out the optimal investment ratio after optimisation.

2. **Input and output specification:**

Inputs:

1. Algorithm parameters such as the number of invested companies, common factor weights, market value thresholds, and number of iterations.
2. Number of Monte Carlo simulations for estimating the Sharpe ratio.

Output:

1. Text output: optimal investment ratio optimal_p.
2. Graphical output: a line graph of the change in investment ratio with the number of iterations, showing the optimisation process of the investment ratio for each company.
3. Table output: export the Sharpe ratio and investment allocation ratio to excel for subsequent confidence interval validation.

3. **Advantages and Disadvantages Analysis**

1. **Advantages:**
   a. The code structure is clear, detailed comments, easy to understand and modify.
   b. Using the SPSA algorithm, gradients can be estimated with a small amount of computation, suitable for high-dimensional problems.
   c. By generating random variables, simulate the market situation, and the model is closer to the reality.
   d. Multiple iterations of dynamic adjustment of the investment ratio, gradually approaching the optimal solution.
   e. Chart changes in Sharpe ratio, expected return, and investment ratio to visualize the optimization process.
2. **Disadvantages**:
   a. Large sample size and multiple iterations result in long running time.
   b. The effect depends on the choice of parameters, and inappropriate parameters may affect the optimization effect.
   c. Independent assumptions about market shocks and firm risks may differ from reality.
   d. It is possible to fall into local optimality, especially in complex non-convex optimization problems.
   e. Focus on Sharpe ratio and fail to consider other risk management strategies such as maximum retracement and value at risk

5.2.2 Project 2

1. **Overview of the programme**

1. **import library:**

   Use numpy for numerical calculations, matplotlib for plotting, and tqdm for displaying the progress bar.

2. **parameter setting:**

   Set the number of customers, the target waiting time quartile, the initial arrival time mean, the initial service time mean, the number of iterations and the number of samples for gradient estimation.

3. **Define the function:**
    a. Decay function for step size and perturbation: control parameter update in SPSA algorithm.
    b. Functions to run individual trajectories: calculate the waiting time sequence for each customer.
    c. Functions for calculating waiting times: calculates the average waiting time and the 90% quantile.
    d. Function to target: calculate the square of the difference between the 90% quantile waiting time and the target value.
4. **SPSA algorithm implementation:**

    Minimise the objective function by iteratively updating the mean service time value so that the 90% quantile of the waiting time is as close as possible to the objective value.
5. **Main Procedure:**
    a. Run the SPSA algorithm to get the history of service time averages.
    b. Find the optimal service time mean and print the results.
    c. Plot the optimisation process for the service time mean.
    d. Generate sample data, calculate the waiting time series, and plot the waiting time distribution.

2. Input and output specification:

Input:

algorithm parameters (e.g., learning rate, number of iterations), investment threshold, number of Monte Carlo simulations, etc.

Output:

history of optimal investment ratio, Sharpe ratio, graphical output of investment ratio change with iteration.

3. Advantages and disadvantages analysis:

1. **Advantages:**
    a. the program implements the SPSA algorithm, which effectively handles the randomness and uncertainty in the optimisation problem.
    b. Using Monte Carlo method to estimate the Sharpe ratio increases the accuracy of the estimation.
    c. Visual output helps to intuitively display the optimisation process and results.
2. **Disadvantages**:
    a. the step size and perturbation parameters in the procedure are fixed and may need to be adjusted to the specific problem.
    b. The procedure does not include validation of the input parameters, which may lead to errors or anomalies in practical applications.
    c. The comments and documentation of the program could be more detailed to allow other developers to quickly grasp the logic of the code.

6. Validation and verification

6.1 Experimental setting

6.1.1 Sample size

The computational budget we used is determined via the total number of updates N that can be sampled. In order to:

1. Minimize the data collection cost
2. Improve the computational efficiency of analysis and simulation
3. Avoid overfitting

We decided to maintain a moderate sample size.

project 1：

However, this sample size still needs to meet several statistical requirements to effectively simulate the maximum Sharp Ratio and optimal allocation vector (p1, p2, p3). Here as follow are the criterias we used to determine whether the sample size is sufficient for simulation:

- Convergence analysis: During the initial trial runs, the convergence of the output indicators can be observed. As the sample size increased to around 1000 and the iteration time went to around 1000, the result converge stably to a certain value.

Therefore we choose 30000 as our total budget, for each run we will have 1000 updates(n = 1000), and we will have 30 parellal runs.

project 2：

In Project 2, we found that 3000 iteration time is enough for convergence, and 30 parallel runs will be sufficient to produce a valid result. So our total budget is still 30000.

6.1.2 The setting of other factors

Project 1:

1. a (Step Size Factor)
    i. Role: Controls the magnitude of the parameter updates in each iteration.
    ii. Reason for Setting: A larger value of □a can accelerate convergence in the initial stages but may cause oscillations or instability. A smaller value allows for finer adjustments in the later stages, ensuring stable convergence. 0.5 is a common initial value that works well in most scenarios.
2. c (Perturbation Factor)
    i. Role: Controls the size of the perturbations used to estimate the gradient in each iteration.
    ii. Reason for Setting: A larger value of c provides better gradient estimates in the initial stages but may introduce more noise. A smaller value offers more precise gradient estimates in the later stages. 0.1 is a typical initial value that balances these needs.
3. A (Stability Factor)
    i. Role: Adjusts the step size factor a to ensure larger steps in the initial iterations and smaller steps as the algorithm progresses.
    ii. Reason for Setting: A is usually set to 1 or close to 1 to ensure that the step size is large enough at the beginning to facilitate rapid convergence and gradually decreases to ensure stable convergence.
4. alpha (Step Size Decay Exponent)
    i. Role: Controls the rate at which the step size factor a decays.

ii. Reason for Setting: The value of □α typically ranges between 0.5 and 1. A value of 0.602 is an empirical choice that ensures the step size decays at a moderate rate, balancing rapid and stable convergence.

5. gamma (Perturbation Decay Exponent)
   i. Role: Controls the rate at which the perturbation factor c decays.
   ii. Reason for Setting: The value of γ usually ranges between 0.1 and 0.2. A value of 0.101 is an empirical choice that ensures the perturbation size decays at a moderate rate, providing good gradient estimates initially and reducing noise later on.

Project 2:

As for the allocation of m and k, we need to keep the total budget m*k fixed. Following are the key considerations:

1) Accuracy of the quantile estimate $W_N(\theta n)$
2) Convergence speed of the SA algorithm

Generally, a higher k (more sub-problem updates) will lead to a more accurate estimation for the quantile $W_N(\theta n)$ at θn, but a slower convergence of the SA algorithm, since m will simultaneously reduce. A higher m (more updates of the SA generating $\{\theta_n\}$) will lead to faster convergence, but the quantile estimates may be less accurate.

Taking these trade-offs into serious consideration, after several trials, we finally found a good allocation of m and k:

m = 1000, k = 30

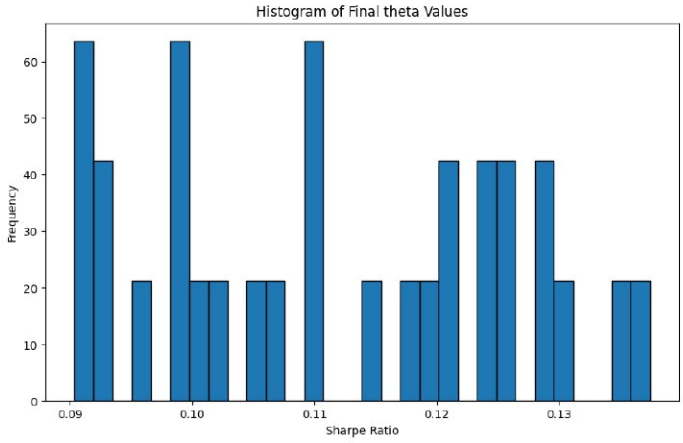
7. Results

7.1 Project 1

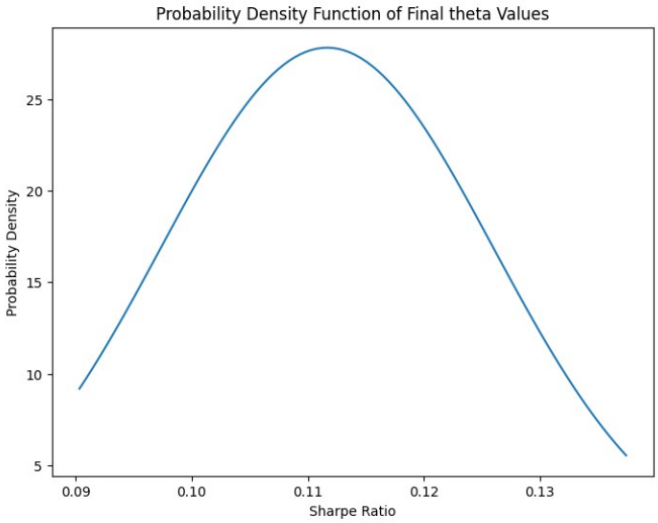
When the initial allocation vector p = (1/3 , 1/3 , 1/3)

For the optimal Sharpe Ratio:

1) Average = 0.110504291
2) Histogram:



3) Probability density function:



4) p-Value:

   By utilizing the Jarque-Bera Test and calculating the p-Value, we found out that the p-Value for Sharpe Ratio's distribution = 0.3400 > 0.05. Therefore we can conclude that the Sharpe Ratio is accord with normal distribution, the result should be valid.

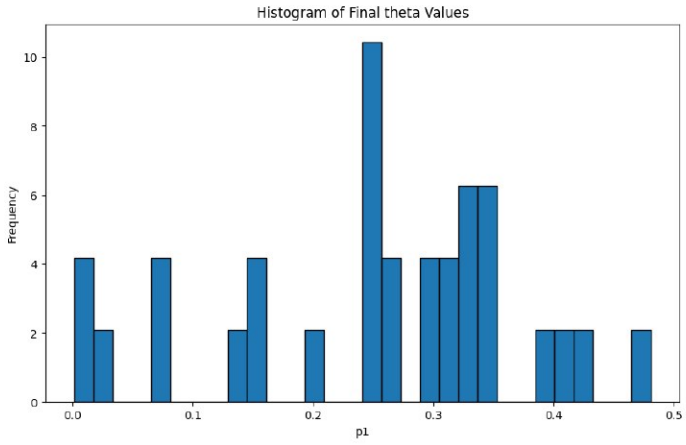
5) Confidence bound:

   95% confidence interval: [0.1062, 0.1171]

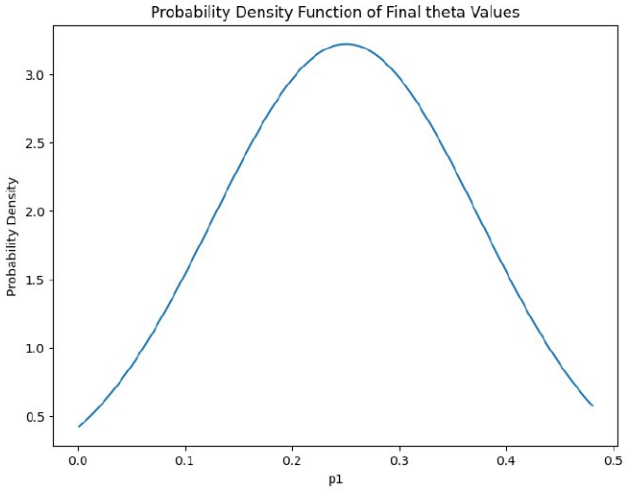
We found out that the p-Value for Sharpe Ratio's distribution is larger than 0.05, therefore we can conclude that the Sharpe Ratio is accord with normal distribution. And the average value of Sharpe Ratio is within its confidence interval. All in all, the result should be valid.

For the optimal allocation vector (p1, p2, p3) :

1) p1:
   a) Histogram



   b) Probability density function:



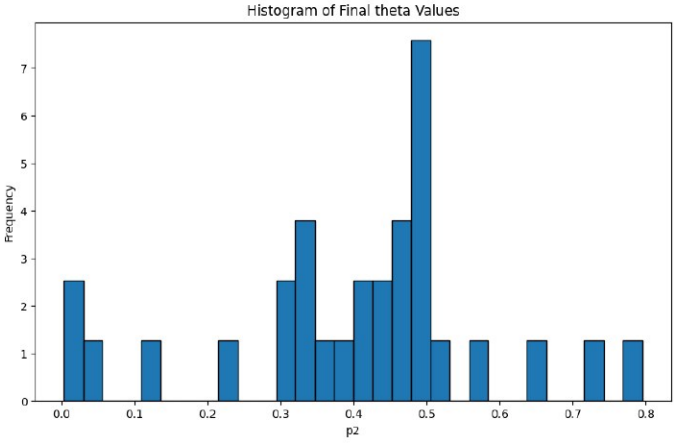
i) By utilizing the Jarque-Bera Test, we calculated that the p-Value for p1's distribution = 0.4946 > 0.05, therefore we can conclude that the value for p1 is accord with normal distribution.

ii) 95% confidence interval: [0.1874, 0.3141]

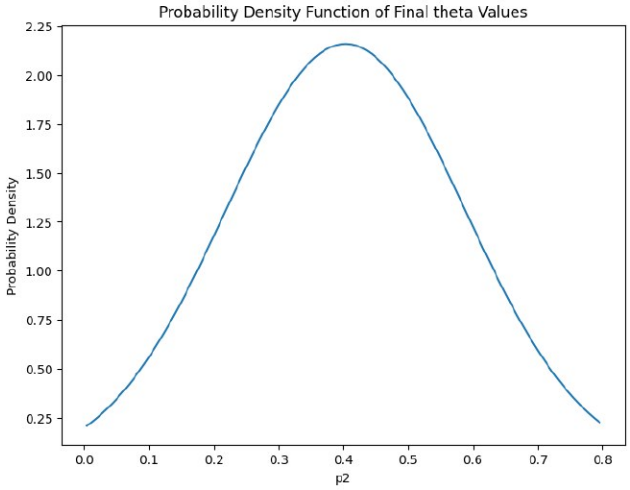
iii) Average p1 = 0.2507

2) p2:

a) Histogram:



b) Probability density function:



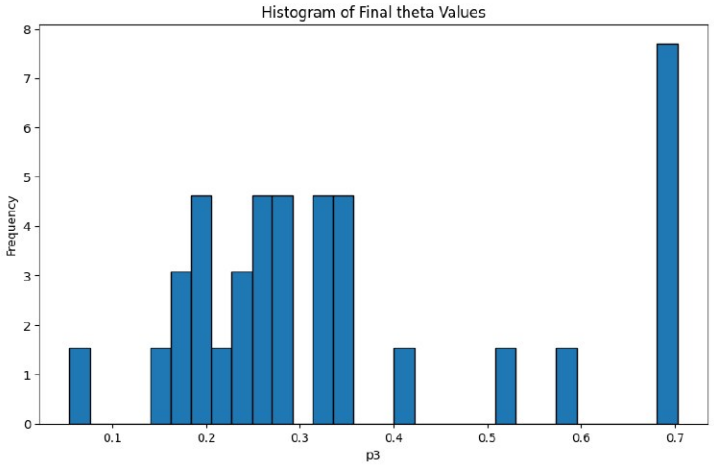
i) p-Value for p2's distribution = 0.5968 > 0.05, therefore we can conclude that the value for p2 is accord with normal distribution.

ii) 95% confidence interval: [0.3086, 0.4979]

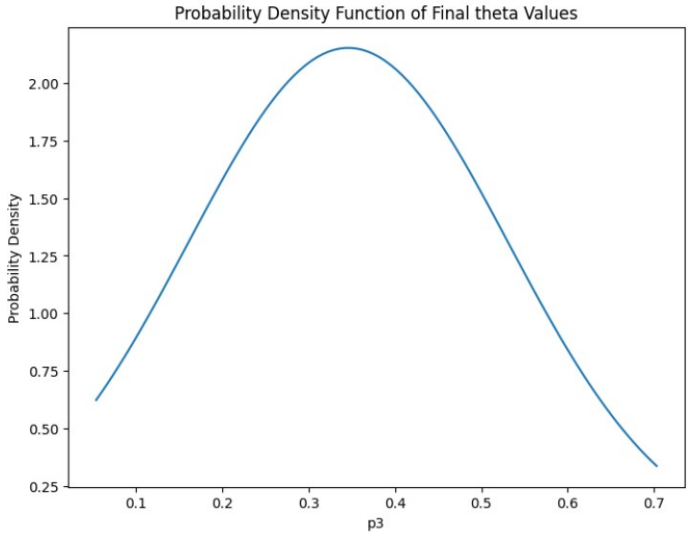
iii) Average p2 = 0.4033

3) p3:

a) Histogram:



b) Probability density function:



i) p-Value for p3's distribution = 0.5968 > 0.05, therefore we can conclude that the value for p3 is accord with normal distribution.

ii) 95% confidence interval:[0.2511, 0.4409]

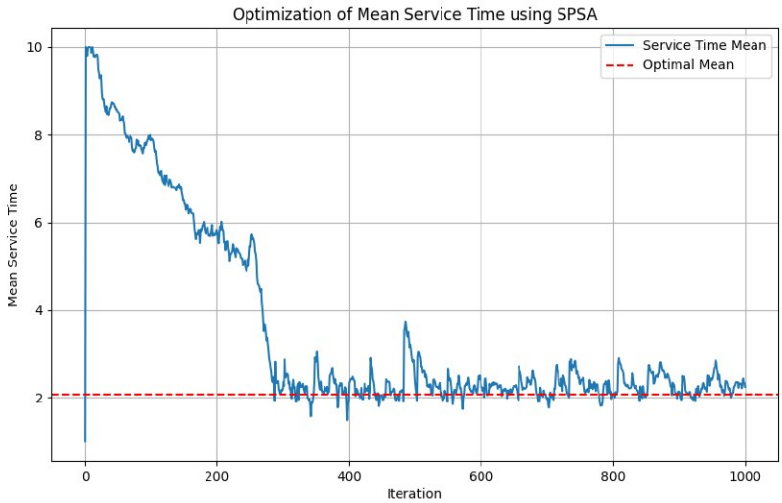
iii) Average p3 = 0.3460

Conclusion:

The optimal allocation vector (p1, p2, p3) is acord with normal distribution, and the Average value of each parameter is within its confidence interval, so that the simulation result should be valid.

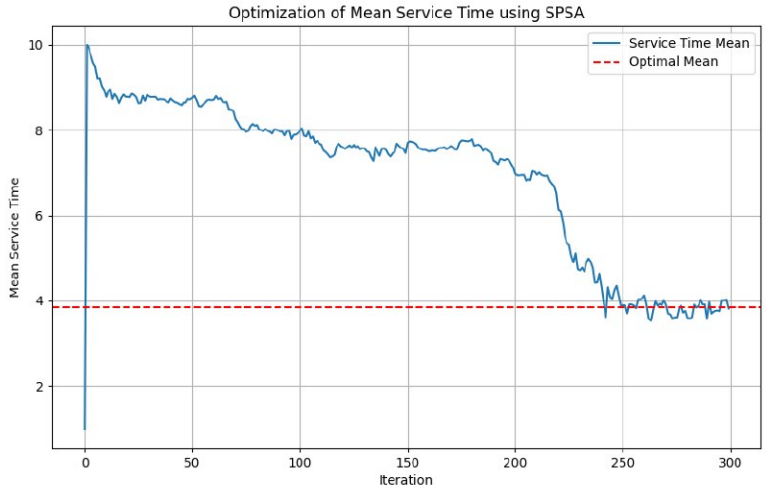
7.2 Project 2:

1) Optimal mean service time
   a) k=30, m=1000



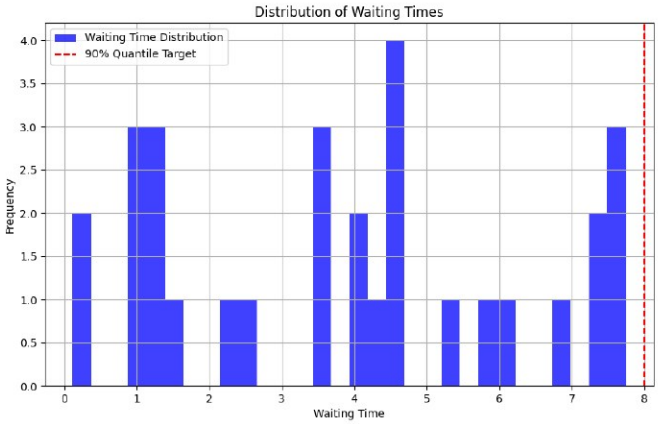
Optimal service time mean: 2.2473526408969304

   b) k=100, m=300



Optimal service tiume mean: 3.8422413957716937

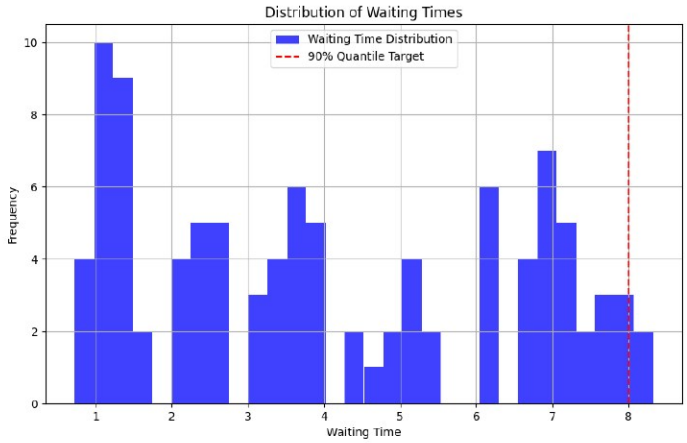
2) Histogram for waiting times in 30 runs:
   a) k=30, m=1000



Jarque-Bera statistic: 1.7590646452288816
p-value: 0.4149769416374659 > 0.05
Acord with normal distribution

95% confidence interval: [2.6917, 4.6036]
Mean waiting time: 3.6476284398149046

   b) k=100, m=300



Jarque-Bera statistic: 8.708594875365213

p-value: 0.012851465368510543

wrong

Final theta value 95% confidence interval: [3.4559, 4.7121]

4.083977216719841

Conclusion:

When iteration times m = 1000 and parallel run times k = 30, the optimal mean service time converges to around 2.064, the distribution of waiting times is normal distribution, and the mean waiting time is within the 95% confidence interval. Therefore the result should be valid and the allocation of m and k is desirable.

8. Interpretation of results

8.1 Project 1

The result shows the optimal proportions of capital allocation among different companies, which have been optimized by the algorithm to maximize the Sharpe ratio. Changes in capital allocations may reflect changes in market conditions, company-specific risks, and expected returns. A high Sharpe ratio may indicate that the investment strategy is efficient in terms of risk control, but it can also mean that the strategy is overly dependent on certain assumptions, such as the accuracy of the market's distribution.

8.2 Project 2

The result illustrates the best expectation of the mean service time driven by the SPSA algorithm. By optimizing the q_0 term, we can minimize operating costs on the premise of satisfying customer service quality. Plus, rationally allocating the value of M and K provided that their product the same can balance the accuracy and running speed, and gradually achieve our demand.

9. Source of errors and difficulties

In this study, we have applied the SPSA algorithm on two different problems in order to provide an in-depth analysis of the investment strategy optimisation and service quality

control problems, respectively. Despite the results achieved, some specific and objective challenges were encountered:

1. **Limitations of model assumptions:** our model is based on specific assumptions, such as assuming independent and homogeneous distribution of firm returns in Project 1, and assuming independent and homogeneous distribution of service time in Project 2. These assumptions simplify the problem but may not adequately capture the complexity of market and customer behaviour, such as market correlation or multi-peaked distributions of service times.
2. **Impact of data quality:** the accuracy of the input data directly affects the simulation results. We face the problem of data noise, missing values and outliers, which can have a direct negative impact on the performance of the algorithm, leading to relative or absolute deviations in the results.
3. **The challenge of algorithm parameter selection:** the convergence of almost all SA algorithms depends on the proper selection of parameters such as step size and perturbation amplitude. Inappropriate parameter settings may lead to slow convergence of the algorithms or failure to converge to the global optimal solution, which is reflected in both Project 1 and Project 2.
4. **Algorithm stability and convergence:** when dealing with problems with a high degree of stochasticity, such as simulating a large number of customer arrivals and service processes, our algorithms require special attention to accommodate this stochasticity and ensure that robust solutions are found.

To address these challenges, we have taken the following steps:

- Conducted sensitivity analyses to assess the impact of key assumptions on model outputs.
- Enhanced the validation process to improve the accuracy and completeness of the data.Implemented parameter tuning to experimentally determine the range of parameters that promote algorithm convergence.

Despite these measures, we recognise that there may still be room for improvement in the model and algorithm. Future work needs to further consider the realistic applicability of the model, the dynamic tuning of the algorithm parameters, and the stability and convergence of the algorithm under a wider range of conditions. Through these efforts, we expect to improve the effectiveness and reliability of the SA algorithm in solving real-world problems.

10. Conclusion

In this study, we successfully apply the SPSA algorithm to solve two practical problems: investment strategy optimisation and service quality control. Project 1 focuses on helping investors allocate capital among three different firms to maximise the risk-adjusted performance (Maximising the Sharpe ratio). Through simulation experiments, we find optimal capital allocation ratios that can be dynamically adjusted to market fluctuations. Project 2, on the other hand, addresses a queuing model for service stations to ensure that the average waiting time of customers does not exceed a predetermined quality control standard. The experimental results show that operational costs can be significantly reduced and service efficiency enhanced by appropriately adjusting the service time parameters. Combining these two projects, the main insight we derive is that dynamic adjustment of strategies is crucial in both financial investment and service management domains. For practitioners, we recommend that investors regularly use the optimisation model in Project 1 to assess market changes and adjust their investment portfolios, while service station managers should flexibly adjust their service strategies during peak hours based on the optimisation results in Project 2 to reduce

customer waiting time and improve service efficiency.

Looking ahead, we believe that the research and application of SA algorithms are promising. Parallelisation of algorithms, adaptivity, multi-stage optimisation, dynamic adjustment of services, generalisation to a wider range of optimisation problems, and integration with machine learning techniques are all directions worth exploring in depth. In addition, the development of user-friendly interfaces and tools for long-term performance evaluation, and the application of optimisation techniques to policy formulation and resource planning are recommended directions for future research. Through these efforts, we can continue to improve the performance and utility of SA algorithms and provide more accurate and efficient solutions for decision makers in various industries.

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12 Appendix

Project code available on request